

Existence and Regularity of Competitive Equilibria in Economies with Externalities

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Abstract

This paper examines the existence and regularity of competitive equilibria in a pure exchange economy with consumption externalities. Using differential topology, we prove that an equilibrium always exists by constructing a continuous homotopy deformation mapping on a compact space. Furthermore, we employ the Transversality Theorem to demonstrate that regular economies form an open and full Lebesgue measure subset of the parameter space. These findings guarantee that the equilibrium set is finite and depends smoothly on underlying economic parameters, providing a rigorous foundation for classical comparative statics under behavioral market frictions.

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1 Introduction

Externalities constitute a fundamental feature of economic interactions, wherein an individual's decisions directly influence other agents' preferences and allocations, often leading to market inefficiencies. In general equilibrium theory, pioneering works by [Laffont \(1978\)](#); [Hammond \(1998\)](#) established that consumption externalities systematically alter market behaviors. While classical frameworks demonstrate that separable preferences can neutralize these external impacts ([Dufwenberg et al. \(2011\)](#)), empirical and theoretical evidence confirms that non-separable behavioral pressures significantly distort baseline equilibrium structures ([Dupor and Liu \(2003\)](#); [Bonnisseau \(2003\)](#)). This

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underscores the critical importance of evaluating both the existence and structural stability of equilibria when socio-economic interactions are present.

To address these interactions, this paper analyzes a smooth pure exchange economy characterized by widespread price-dependent and endowment-regarding consumption externalities. Each consumer's preference is represented by a utility function simultaneously affected by the allocations, prices, and initial endowments of others. In this setting, the competitive equilibrium is formalized as an equilibrium à la Nash. The central objective of this study is to guarantee the generic regularity of such economies. An economy is deemed regular if it possesses a finite, odd number of equilibria, with each equilibrium exhibiting local, differentiable dependence on the underlying parameters. Such regular structures are essential for validating classical comparative statics (Mas-Colell (1985); Balasko (2016)). Despite extensive research on external effects (Bonnisseau (2003); Bonnisseau and Del Mercato (2010)), a significant research gap persists regarding the generic regularity of economies that incorporate complex consumption constraints alongside multi-dimensional externalities. Existing literature often relies heavily on restrictive geometric assumptions on preferences (Bonnisseau (2003)) or focuses exclusively on simplified consumption dynamics (Balasko (2015)). Furthermore, traditional existence proofs predominantly rely on standard fixed-point arguments (Arrow (1971); Laffont and Laroque (1972)).

This paper bridges these gaps by employing advanced differential topology techniques instead of fixed-point methods. Our contribution is two-fold: first, we prove the global existence of competitive equilibria by constructing a continuous homotopy deformation mapping on a compactified endogenous space. Second, by introducing a robust boundary condition and ensuring that own-consumption effects dominate aggregate external variations, we apply the Transversality Theorem to demonstrate that regular economies form an open and full Lebesgue measure subset of the parameter space. These findings mathematically guarantee that the equilibrium set is strictly finite, structurally stable, and depends smoothly (C^1) on economic parameters, establishing a rigorous micro-foundation for policy analysis under external market frictions.

The paper is organized as follows. In Section 2, we present the model and the assumptions. In Section 3, we provide the notions of competitive equilibrium and equilibrium function.

2 The exchange economy with wealth concern

In this section, we describe the basic assumptions and definition regarding the exchange model with wealth externalities.

2.1 Goods and price

We consider a model with a finite number L of distinguishable goods; these are indicated by an index l running from 1 to L . Here commodity space is $\mathbb{R}^L$, the whole L -dimension Euclidean space. A commodity bundle or commodity basket is a vector $x = (x^1, \dots, x^L) \in \mathbb{R}^L$. Component x^l represents the quantity of commodity l in the commodity basket x . Let $x^\backslash = (x^1, \dots, x^{L-1}) \in \mathbb{R}^{L-1}$ is obtained from x by deleting the last coordinate x^L .

Each commodity l has a price p_l , which is a real number. The price p_l can be either positive, negative, or null. Here we consider the scarce commodity, then the prices are strictly positive. The price system or price vector is $p = (p_1, \dots, p_L) \in \mathbb{R}_{++}^L$. Often, one considers normalized price vector. Different normalizations are possible. The default assumption is to normalize prices by choosing commodity L as *numéraire*, which means that its price p_L is fixed to 1. The set of numéraire normalized prices is denoted by $\mathbb{S} = \mathbb{R}_{++}^{L-1} \times \{1\}$. The numéraire normalized price vector $p \in \mathbb{S}$ can be written as $p = (p^\backslash, 1)$ where $p^\backslash = (p_1, \dots, p_{L-1}) \in \mathbb{R}_{++}^{L-1}$ expresses the prices of the non-numéraire goods.

We denote $P(\mathbb{S})$ the closure of the set $\mathbb{S}$. Then $P(\mathbb{S})$ is a closed set of the real projective space $\mathbb{P}_L(\mathbb{R})$, which is the set of lines through the origin of space $\mathbb{R}^L$. More precisely, $(p_1, p_2, \dots, p_L)$ and $(\alpha p_1, \alpha p_2, \dots, \alpha p_L)$ where $\alpha \in \mathbb{R} \setminus \{0\}$, represent the same point in $P(\mathbb{S})$. Since $\mathbb{P}_L(\mathbb{R})$ is compact, the closure set $P(\mathbb{S})$ is compact.

2.2 Endowments and wealth

There is also a finite number m of consumers, and each one of them is indicated by an index $i = 1, \dots, m$. The consumption set or consumption space is a subset X of $\mathbb{R}^L$, which contains all possible commodity bundles. We assume that the consumption space of every consumer is the full commodity space $X = \mathbb{R}^L$. Consumer i , with $1 \leq i \leq m$, has the initial endowment vector $e_i \in X$. Given the price vector $p \in \mathbb{S}$, consumer i 's wealth w_i is the inner product $w_i = p \cdot e_i$.

Let $\Omega = X^m$ be the endowment or parameter space. The m -tuple $e = (e_1, \dots, e_m) \in \Omega$ represents the endowments of all the consumers in the economy. The total resources $r = \sum_i e_i$ are variable. The vector obtained from $e = (e_1, \dots, e_i, \dots, e_m) \in \Omega$ by deleting the i -th component e_i is denoted by $e_{-i} = (e_1, \dots, e_{i-1}, e_{i+1}, \dots, e_m) \in \Omega$. Similarly, $e_{-i}^\backslash$ (resp. w_{-i}) is obtained from $e^\backslash = (e_1^\backslash, \dots, e_m^\backslash)$ (resp. $w = (w_1, \dots, w_m)$) by deleting the i -th component $e_i^\backslash$ (resp. w_i).

We define $\mathbb{B} = \mathbb{S} \times X^{m-1} \times \mathbb{R}$ the set of variables. This set is obviously diffeomorphic to Euclidean space $\mathbb{R}^{Lm}$. The vector $b = (p, e_1, \dots, e_{m-1}, w_m) \in \mathbb{B}$

describes the prices of the L goods, the endowment of $m - 1$ consumers except consumer m and the wealth of consumer m .

2.3 Utility functions and basic assumptions

Each consumer has a preference represented by an utility function. The utility function of consumer i is a function $u_i : \mathbb{R}_{++}^{LI} \times \mathbb{S} \times \Omega \rightarrow \mathbb{R}$ that associates the consumer i 's utility level $u_i(x, p, e)$ with the allocation x , the price vector p and the endowment vector $e = (e_1, \dots, e_m)$. The economy now is $\mathcal{E} = (u_i, e_i)_{i=1,2,\dots,m}$ described by initial endowments and m -tuple of utility functions.

Let S be the unit simplex of $\mathbb{R}^L$. We firstly extend below some classical assumptions on utility functions to our exchange model.

Assumption 1 For all $i = 1, 2, \dots, I$

- (1) u_i is $\mathcal{C}^3$ on $\mathbb{R}_{++}^{LI} \times \mathbb{S} \times \Omega$.
- (2) For each $(x_{-i}, p, e) \in \mathbb{R}_{++}^{L(I-1)} \times \mathbb{S} \times \Omega$, the function $u_i(\cdot, x_{-i}, p, e)$ is differentiable strictly increasing, i.e., for every $x_i \in \mathbb{R}_{++}^L$, $D_{x_i} u_i(x_i, x_{-i}, p, e) \in \mathbb{R}_{++}^L$.
- (3) For each $(x_{-i}, p, e) \in \mathbb{R}_{++}^{L(I-1)} \times \mathbb{S} \times \Omega$, the function $u_i(\cdot, x_{-i}, p, e)$ is differentiable strictly quasi-concave, i.e., for every $x_i \in \mathbb{R}_{++}^L$, $D_{x_i}^2 u_i(x_i, x_{-i}, p, e)$ is negative definite on $\text{Ker } D_{x_i} u_i(x_i, x_{-i}, p, e)$.

The assumptions above are usually made in a classical smooth exchange economies.¹

As is well known, in the pure exchange model with externalities, to control the behavior at the boundary for proving compactness property needed for the existence of equilibrium and regularity result, one need a boundary assumption.²

Let S be the unit simplex of $\mathbb{R}^L$. For all $i = 1, 2, \dots, I$, consider a mapping $\tilde{u}_i$ from $\mathbb{R}_{++}^{LI} \times (S \cap \mathbb{R}_{++}^L) \times \Omega$ to $\mathbb{R}$ defined by $\tilde{u}_i(x, q, e) = u_i(x, (1/q^L)q, e)$. Note that $\tilde{u}_i$ and the derivative with respect to x_i of $\tilde{u}_i$ is $D_{x_i} \tilde{u}_i(x, q, e) = D_{x_i} u_i(x, (1/q^L)q, e)$.

Assumption 2 For all $i = 1, 2, \dots, I$, there exists a differentiable extension of $\tilde{u}_i$ on $\mathbb{R}_{++}^L \times \mathbb{R}_+^{L(I-1)} \times S \times \Omega$ such that for each $(x_i, x_{-i}, q, e) \in \mathbb{R}_{++}^L \times \mathbb{R}_+^{L(I-1)} \times S \times \Omega$,

¹ See for example Villanacci et al. (2013) and Del Mercato (2006).

² See for example Del Mercato and Platino (2017) and Nguyen (2021).

- (i) $cl_{\mathbb{R}^L} \{x'_i \in \mathbb{R}_{++}^L : \tilde{u}_i(x'_i, x_{-i}, q, r) \geq \tilde{u}_i(x_i, x_{-i}, q, r)\} \subseteq \mathbb{R}_{++}^L$;
- (ii) $D_{x_i} \tilde{u}_i(x, q, r) \in \mathbb{R}_{++}^L$.

Points (ii) simply means that the extension of the utility function is strictly increasing, and point (i) implies that if the consumption of some commodities becomes very small, then the amount of some other commodities have to become very large in order to remain the same utility level.

Besides those standard assumptions, we consider an additional assumption to ensure the genericity of regular economies. The set of regular economies makes a crucial role in this paper.

Assumption 3 *Let $(x, e, v, p) \in \mathbb{R}_{++}^{Lm} \times \Omega \times \mathbb{R}^{Lm} \times \mathbb{S}$. If for all consumer i , $v_i \in \text{Ker } D_{x_i} u_i(x_i, x_{-i}, p, e)$ and the gradient $D_{x_i} u_i(x_i, x_{-i}, p, e)$ is proportional to the prices vector p , then the following property is satisfied for all consumers $k = 1, \dots, m$.*

$$v_k D_{e_k x_k}^2 u_k(x_k, x_{-k}, p, e)(v_k) + v_k D_{x_k}^2 u_k(x_k, x_{-k}, p, e)(v_k) < 0 \text{ whenever } v_k \neq 0, \\ (v_i D_{x_k x_i}^2 u_i(x_i, x_{-i}, p, e) + v_i D_{e_k x_i}^2 u_i(x_i, x_{-i}, p, e))(v_k) = 0 \text{ for all } i \neq k.$$

Assumption 3 is in the same spirit as Assumption 10 in [Bonnisseau and Del Mercato \(2010\)](#) and [Nguyen \(2021\)](#). The idea is that the effect of the own consumption dominates the aggregate of the second order of the consumption's external effect. Moreover, this assumption also puts some restrictions on the first-order effect of consumption externalities; see [Nguyen \(2021\)](#) for more details. Remark that this assumption is satisfied in the economy without externalities and in the model in [Geanakoplos and Polemarchakis \(2008\)](#) where they consider additively separable utility functions.³

We now denote $\mathcal{U}$ the set of all I -tuple utility functions $(u_i)_{i=1, \dots, I}$ which satisfies Assumptions 1, 2 and 3. We endow the space $\mathcal{U}$ with the topology of $\mathcal{C}^3$ uniform convergence on compact sets.

3 Competitive equilibrium

For every $y = (y^1, \dots, y^L) \in \mathbb{R}^L$, we denote $y^\setminus = (y^1, \dots, y^{L-1}) \in \mathbb{R}^{L-1}$.

In a differential setting, we characterize the equilibria by a system of first-order conditions and market clearing conditions. Following works by [Smale \(1974\)](#), [Villanacci et al. \(2013\)](#), we define the set of endogenous variables as $\Xi :=$

³ Indeed, in these two models, $D_{x_i x_k}^2 u_k(x_k, x_{-k}, p) = 0$ for all i and k .

$\mathbb{R}_{++}^{LI} \times \mathbb{R}_{++}^I \times \mathbb{R}_{++}^{L-1}$, with a generic element $\xi := (x, \lambda, p^\setminus) := ((x_i, \lambda_i)_{i=1,2,\dots,I}, p^\setminus)$. Under Assumption 1, the equilibrium is defined as follow.

Definition 1 (x^*, p^*) is an competitive equilibrium if there exists $\lambda^* \in \mathbb{R}_{++}^I$ such that $F_e(\xi^*) = 0$ where the equilibrium function $F_e : \Xi \rightarrow \mathbb{R}^{\dim \Xi}$ is defined by:

$$F_e(\xi) = \left((F_e^{(i,1)}(\xi), F_e^{(i,2)}(\xi))_{i=1,2,\dots,I}, F_e^M(\xi) \right).$$

where for each $i = 1, 2, \dots, I$

$$F_e^{(i,1)}(\xi) := D_{x_i} u_i(x_i, x_{-i}, p, e) - \lambda_i p$$

$$F_e^{(i,2)}(\xi) := -p \cdot (x_i - e_i)$$

$$\text{and } F_e^M(\xi) := \sum_{i=1}^I x_i^\setminus - \sum_{i=1}^I e_i^\setminus.$$

We recall below the definition of regular economies.

Definition 2 $e \in \Omega$ is a regular economy if 0 is a regular value of the mapping F_e , that is for each $\xi^* \in F_e^{-1}(0)$, the differential mapping $D_\xi F_e(\xi^*)$ is onto.

We denote $\mathcal{R}^u$ the set of regular economies associated to the profile of utility functions u . We now present the existence of equilibrium and the regularity results in this differentiable setting.

3.1 Existence result

Theorem 1 Equilibrium exists for any $(u, e) \in \mathcal{U} \times \Omega$.

Proof of Theorem 1. For given $q^* \in \mathbb{S}$ and $x^{**} = (x_1^{**}, \dots, x_m^{**}) \in \mathbb{R}_{++}^{Lm}$, we define the function $L : \Xi \rightarrow \mathbb{R}^{\dim \Xi}$ as follow

$$L(\xi) := \begin{pmatrix} D_{x_1} u_1(x_1, e_{-1}, q^*, e) - \lambda_1 p \\ -p \cdot (x_1 - x_1^{**}) \\ \vdots \\ D_{x_m} u_m(x_m, e_{-m}, q^*, e) - \lambda_m p \\ -p \cdot (x_m - x_m^{**}) \\ \sum_{i=1}^m x_i^\setminus - \sum_{i=1}^m x_i^{**\setminus} \end{pmatrix}$$

Note that under Assumption 1, $L(\xi) = 0$ has unique solution, $L^{-1}(0) = \{\xi^* = ((x_i^{**}, \lambda_i^{**})_{i=1,\dots,m}, p^{**\setminus})\}$ and $D_\xi L(\xi^{**})$ has full rank because e and q^* are given

as constants. The utility function $u_i(x_i, e_{-i}, q^*, e)$ is then considered as a standard utility function without externalities. Then the existence and uniqueness of the allocation x^{**} comes from the standard argument for the existence and uniqueness of a Pareto optimal allocation as in Villanacci et al. (2013).

We now consider the homotopy $H : \Xi \times [0, 1] \rightarrow \mathbb{R}^{dim\Xi}$:

$$H(\xi, t) := \begin{pmatrix} D_{x_1} u_1(x_1, tx_{-1} + (1-t)e_{-1}, tp + (1-t)q^*, e) - \lambda_1 p \\ -p \cdot (x_1 - (te_1 + (1-t)x_1^{**})) \\ \vdots \\ D_{x_m} u_m(x_m, tx_{-m} + (1-t)e_{-m}, tp + (1-t)q^*, e) - \lambda_m p \\ -p \cdot (x_m - (te_m + (1-t)x_m^{**})) \\ \sum_{i=1}^m x_i^\setminus - \sum_{i=1}^m (te_i^\setminus + (1-t)x_i^{**\setminus}) \end{pmatrix}$$

Then $H(\xi, 0) = L(\xi)$ and $H(\xi, 1) = \tilde{F}_e(\xi)$. We denote

$$\begin{aligned} H^{(i,1)}(\xi, t) &= D_{x_i} u_i(x_i, tp + (1-t)q^*, e) - \lambda_i p, \\ H^{(i,2)}(\xi, t) &= -p \cdot (x_i - (te_i + (1-t)x_i^{**})), \\ H^M &= \sum_{i=1}^m x_i^\setminus - \sum_{i=1}^m (te_i^\setminus + (1-t)x_i^{**\setminus}), \end{aligned}$$

and

$$\begin{aligned} e_i^n &= t^n e_i + (1-t^n)x_i^{**}, \\ p_t^n &= t^n p^n + (1-t^n)q^* \in \mathbb{S}, \\ q_t^n &= (1/\sum_{l=1}^L p_t^{ln})p_t^n \in S \cap \mathbb{R}_{++}^L. \end{aligned}$$

We prove that $H^{-1}(0)$ is compact. Take any sequence $(\xi^n, t^n)_{n \in \mathbb{N}} \subset H^{-1}(0)$, we prove that there exists a subsequence converging to a point in $H^{-1}(0)$. It is clear that, up to a subsequence, $(t^n)_{n \in \mathbb{N}}$ (resp., $(q_t^n)_{n \in \mathbb{N}}$) converges to $t^* \in [0, 1]$ (resp., $q_t^* \in S$) and so do $(e_i^n)_{n \in \mathbb{N}}$.

Claim 1. $(x^n)_{n \in \mathbb{N}}$ admits a subsequence converging to $(x^*) \in \mathbb{R}_{++}^{Lm}$.

From $H^M(\xi^n, t^n) = 0$ and $H^{(i,2)}(\xi^n, t^n) = 0$, we have $\sum_i x_i^n = \sum_i (t^n e_i + (1-t^n)x_i^{**})$. Since t^n converges to t^* , e_i^n converges to $e_i^* = t^* e_i + (1-t^*)x_i^{**}$. Therefore, $(t^n e_i + (1-t^n)x_i^{**})$ is bounded above, which follows $\sum_i (t^n e_i + (1-t^n)x_i^{**})$ is bounded above. Combining with x^n is bounded below by 0, x^n is bounded. Then, up to a subsequence, $(x^n)_{n \in \mathbb{N}}$ converges to $(x^*) \in \mathbb{R}_+^{Lm}$.

From the definition of x^n , we have for any n and any i ,

$$u_i(x_i^n, t^n x_{-i}^n + (1 - t^n)e_{-i}, p_t^n, e) \geq u_i(e_i^n, t^n x_{-i}^n + (1 - t^n)e_{-i}, p_t^n, e)$$

Define $\mathbf{1} := (1, 1, \dots, 1) \in \mathbb{R}_{++}^L$. From Point 2 of Assumption 1, we have for each $\epsilon > 0$

$$u_i(x_i^n + \epsilon \mathbf{1}, t^n x_{-i}^n + (1 - t^n)e_{-i}, p_t^n, e) > u_i(e_i^n, t^n x_{-i}^n + (1 - t^n)e_{-i}, p_t^n, e).$$

So, from the definition of $\tilde{u}_i$, we have

$$\tilde{u}(x_i^n + \epsilon \mathbf{1}, t^n x_{-i}^n + (1 - t^n)e_{-i}, q_t^n, e) > \tilde{u}(e_i^n, t^n x_{-i}^n + (1 - t^n)e_{-i}, q_t^n, e).$$

Taking the limit on n , since (x_i^n, q_t^n, e_i^n) converges to $(x_i^*, q_t^*, e_i^*) \in \mathbb{R}_+^{Lm} \times S \times \mathbb{R}_{++}^L$, and $\tilde{u}_i$ is continuous, we get

$$\tilde{u}(x_i^* + \epsilon \mathbf{1}, t^* x_{-i}^* + (1 - t^*)e_{-i}, q_t^*, e) \geq \tilde{u}(e_i^*, t^* x_{-i}^* + (1 - t^*)e_{-i}, q_t^*, e).$$

for any $\epsilon > 0$. Consequently, $x_i^* \in \mathbb{R}_{++}^L$ since it belongs to the set $cl_{\mathbb{R}^L}\{x_i \in \mathbb{R}_{++}^L : \tilde{u}(x_i, t^* x_{-i}^* + (1 - t^*)e_{-i}, q_t^*) \geq \tilde{u}(e_i^*, t^* x_{-i}^* + (1 - t^*)e_{-i}, q_t^*)\} \subset \mathbb{R}_{++}^L$ by Assumption 2.

Claim 2. $(\lambda^n)_{n \in \mathbb{N}}$ admits a subsequence converging to $\lambda^* \in \mathbb{R}_{++}^m$.

Since for all $n \in \mathbb{N}$ $p^{n,L} = 1$ and $H^{(i,1)}(\xi^n, t^n) = 0$, we have

$$D_{x_i}^L u_i(x_i^n, t^n x_{-i}^n + (1 - t^n)e_{-i}, p_t^n, e) = D_{x_i}^L \tilde{u}_i(x_i^n, t^n x_{-i}^n + (1 - t^n)e_{-i}, q_t^n, e) = \lambda_i^n.$$

By taking the limit on both sides, $D_{x_i}^L \tilde{u}_i(x_i^*, t^* x_{-i}^* + (1 - t^*)e_{-i}, q_t^*, e) = \lambda_i^* > 0$. The strict inequality comes from Assumption 2.

Claim 3. $(p^{n\setminus})_{n \in \mathbb{N}}$ admits a subsequence converging $p^{*\setminus} \in \mathbb{R}_{++}^{L-1}$.

Again from $F^{(i,1)}(\xi^n, t^n) = 0$, $D_{x_i}^{\setminus} u_i(x_i^n, t^n x_{-i}^n + (1 - t^n)e_{-i}, p_t^n, e) = D_{x_i}^{\setminus} \tilde{u}_i(x_i^n, t^n x_{-i}^n + (1 - t^n)e_{-i}, q_t^n, e) = \lambda_i^n p^{n\setminus}$. Therefore, taking the limit, we get

$$\begin{aligned} \lim_{n \rightarrow +\infty} p^{n\setminus} &= \lim_{n \rightarrow +\infty} \frac{D_{x_i}^{\setminus} \tilde{u}_i(x_i^n, t^n x_{-i}^n + (1 - t^n)e_{-i}, q_t^n, e)}{\lambda_i^n} \\ &= \frac{D_{x_i}^{\setminus} \tilde{u}_i(x_i^*, t^* x_{-i}^* + (1 - t^*)e_{-i}, q_t^*, e)}{\lambda_i^*} = p^{*\setminus}. \end{aligned}$$

and $p^{*\setminus} \in \mathbb{R}_{++}^{L-1}$ from Assumption 2.

Finally, since $(\xi^n, t^n)_{n \in \mathbb{N}} \subset H^{-1}(0)$ and H is continuous, we get $H(\xi^*, t^*) = 0$, i.e., $(\xi^*, t^*) \in H^{-1}(0)$, which implies that $H^{-1}(0)$ is compact.

Therefore, applying the Homotopy theorem, $\tilde{F}_e^{-1}(0)$ is compact and non-empty, which ends the proof. ■

3.2 Regularity result

We consider the new function $F : \Xi \times \Omega \rightarrow \mathbb{R}^{\dim \Xi}$ as $F(\xi, e) := F_e(\xi)$. We call $e \in \Omega$ is an extended regular economy if 0 is regular value for mapping F_e . Precisely, $e \in \Omega$ is an extended regular economy if for each $\xi^* \in F_e^{-1}(0)$, the differential mapping $D_\xi F_e(\xi^*)$ is onto. Denote the set of extended regular economies is $\mathcal{R}^*$.

Lemma 1 *0 is a regular value for $F(\xi, e)$*

Proof of Lemma 1. We have to show that for each $(\xi^*, e^*) \in F^{-1}(0)$, the Jacobian matrix $D_{\xi, e} F(\xi^*, e^*)$ is onto, i.e., has full row rank.

Let $\Delta := (\Delta x_i, \Delta \lambda_i)_{i=1,2,\dots,m}, \Delta p^\setminus) \in \mathbb{R}^{(L+1)m} \times \mathbb{R}^{L-1}$. Our target is to show that $\Delta \cdot D_{\xi, e} F(\xi^*, e^*) = 0$ implies $\Delta = 0$. We now write in detail this system.

$$\begin{cases} \sum_{i=1}^m \Delta x_i D_{x_k x_i}^2 u_k(x_i^*, x_{-i}^*, p^*, e^*) - \Delta \lambda_h p^* + \Delta p^\setminus [I_{L-1} | 0] = 0, \forall k & (1) \\ -\Delta x_k \cdot p^* = 0, \forall k & (2) \\ \sum_{i=1}^m \Delta x_i D_{e_k x_i}^2 u_i(x_i^*, x_{-i}^*, p^*, e^*) + \Delta \lambda_h p^* - \Delta p^\setminus [I_{L-1} | 0] = 0, \forall k. & (3) \end{cases}$$

Adding Equations (1) and (3) for k then multiplying with Δx_k , we get

$$\sum_{i=1}^m \left(\Delta x_i D_{x_k x_i}^2 u_k(x_i^*, x_{-i}^*, p^*, e^*) + \Delta x_i D_{e_k x_i}^2 u_i(x_i^*, x_{-i}^*, p^*, e^*) \right) (\Delta x_k) = 0 \quad (4)$$

From Equation (2), Δx_i is orthogonal to p^* , so $\Delta x_i \in \text{Ker } D_{x_i} u_i(x_i^*, p^*, e^*)$ for all $i = 1, 2, \dots, m$. Thus from Assumption 3,

$$\left(\Delta x_i D_{x_k x_i}^2 u_k(x_i^*, x_{-i}^*, p^*, e^*) + \Delta x_i D_{e_k x_i}^2 u_i(x_i^*, x_{-i}^*, p^*, e^*) \right) (\Delta x_k) = 0 \text{ for all } i \neq k.$$

So Equation 4 becomes

$$\left(\Delta x_k D_{x_k}^2 u_k(x_i^*, x_{-i}^*, p^*, e^*) + \Delta x_k D_{e_k x_k}^2 u_i(x_i^*, x_{-i}^*, p^*, e^*) \right) (\Delta x_k) = 0 \quad (5)$$

If $\Delta x_k \neq 0$ then we get contradiction to Assumption 3. Thus, we get $\Delta x_k = 0$ for all k .

Equation (1) for k implies $\Delta \lambda_k = 0$ and also $\Delta p^\setminus = 0$. So $\Delta = 0$. ■

Directly from Lemma 1 and the Regular Value Theorem, we have the following result.

Proposition 1 *$F^{-1}(0)$ is a smooth submanifold of $\Xi \times \Omega$ of dimension Lm .*

For proving the genericity property, we need following lemma.

Lemma 2 *The projection mapping $\Pi : F^{-1}(0) \rightarrow \Omega$, $(\xi, e) \mapsto e$ is proper.*

Proof.

We will prove that any sequence $(\xi^n, e^n)_{n \in \mathbb{N}} \subset F^{-1}(0)$, up to a subsequence, converges to an element in $F^{-1}(0)$, knowing that sequence $(e^n)_{n \in \mathbb{N}}$ converges to $e^* \in \Omega$. We recall that $\xi^n = (x^n, \lambda^n, p^{n\setminus})$ where $x = (x_1, \dots, x_m) \in \mathbb{R}_{++}^{Lm}$, $\lambda = (\lambda_1, \dots, \lambda_m) \in \mathbb{R}_{++}^m$ and $p^\setminus \in \mathbb{R}_{++}^{L-1}$.

- $(x^n)_{n \in \mathbb{N}}$ admits a subsequence converging to $x^* \in \mathbb{R}_{++}^{Lm}$.

From $F^{(i,2)}(\xi^n, e^n) = 0$ and $F^M(\xi^n, e^n) = 0$, $x_j^n = \sum_i e_i^n - \sum_{i \neq j} x_i^n \leq \sum_i e_i^n$

for each $j = 1, \dots, m$. Then $(x^n)_{n \in \mathbb{N}}$ is bounded from above since $(e^n)_{n \in \mathbb{N}}$ converges. And since $(x^n)_{n \in \mathbb{N}}$ is bounded from below by zero, $(x^n)_{n \in \mathbb{N}}$, up to a subsequence, converges to $x^* \geq 0$. Moreover, by definition of x^n , we have $u_i(x_i^n, e_{-i}^n) \geq u_i(e_i^n, e_{-i}^n)$ for any n and any i . Define $\mathbf{1} := (1, 1, \dots, 1) \in \mathbb{R}_{++}^L$, from the point 2 of Assumption ?? we have that for each $\epsilon > 0$, $u_i(x_i^n + \epsilon \mathbf{1}, e_{-i}^n) \geq u_i(e_i^n, e_{-i}^n)$. So taking the limit on n , since (x_i^n, e_i^n) converges to (x_i^*, e_i^*) , and u_i are continuous function, we get $u_i(x_i^*, e_{-i}^*) \geq u_i(e_i^*, e_{-i}^*) := \underline{u}_i$ for any i . By point 4 of Assumption ??, $x_i^* \in \mathbb{R}_{++}^L$ since x_i^* belongs to the set $cl_{\mathbb{R}^L} \{x_i \in \mathbb{R}_{++}^L : u_i(x_i, e_{-i}^*) \geq \underline{u}_i\}$.

- $(\lambda^n)_{n \in \mathbb{N}}$ admits a subsequence converging to $\lambda^* \in \mathbb{R}_{++}^m$.

Since $\forall n \in \mathbb{N} p_L^n = 1$ and $F^{(i,2)}(\xi^n, e^n) = 0$, we have $D_{x_i^L} u_i(x_i^n, e_{-i}^n) = \lambda_i^n$. Taking limit on n on both sides, we have $D_{x_i^L} u_i(x_i^*, e_{-i}^*) = \lambda_i^* > 0$, the strict inequality comes from point 2 of Assumption ??.

- $(p^{n\setminus})_{n \in \mathbb{N}}$ admits a subsequence converging to $p^{*\setminus} \in \mathbb{R}_{++}^{L-1}$.

Again from $F^{(i,2)}(\xi^n, e^n) = 0$, $D_{x_i^\setminus} u_i(x_i^n, e_{-i}^n) = \lambda_i^n p^{n\setminus}$. Therefore, taking the limit, we get

$$\lim_{n \rightarrow +\infty} p^{n\setminus} = \lim_{n \rightarrow +\infty} \frac{D_{x_i^\setminus} u_i(x_i^n, e_{-i}^n)}{\lambda_i^n} = \frac{D_{x_i^\setminus} u_i(x_i^*, e_{-i}^*)}{\lambda_i^*} = p^{*\setminus}.$$

where the strict inequality comes from point 2 of Assumption ??.

Finally, since $(\xi^n, e^n)_{n \in \mathbb{N}} \subset F^{-1}(0)$ and F is continuous, we get $F(\xi^*, e^*) = 0$, i.e, $(\xi^*, e^*) \in F^{-1}(0)$, which finish our proof. ■

We now can prove the important theorem of this subsection.

Theorem 2 *The set of extended regular economies $\mathcal{R}^*$ is an open and full*

Lebesgue measure subset of Ω .

Proof. From Lemma 1, 0 is regular value for mapping F . Then the Transversality Theorem implies that there exists Ω^* full measure subset of Ω such that $\forall e \in \Omega^*$, 0 is regular value of F_e . Therefore, the set of extended regular economies $\mathcal{R}^*$ is full measure since it contains Ω^* .

Define $G = \{(\xi^*, e^*) \in \Xi \times \Omega, (\xi^*, e^*) \in F^{-1}(0) \mid \det D_{\xi, e} F(\xi^*, e^*) = 0\}$. Then we can write $\mathcal{R}^* = \Omega \setminus \Pi(G)$ where Π is defined in Lemma 2. The set G is closed set since the determinant is a continuous function. Therefore, $\Pi(G)$ is closed set by properness of Π , which implies the set $\mathcal{R}^*$ is open. ■ In addition, by directly applying the Implicit Function Theorem and the Regular Value Theorem, we obtain several important properties

Corollary 1 *For given $(u, e) \in \mathcal{U} \times \mathcal{R}_u$, we have*

- 1 *there exists $r \in \mathbb{N} \setminus \{0\}$ such that $F_e^{-1}(0) = \{\xi_1, \dots, \xi_r\}$;*
- 2 *there exist an open neighborhood V of e in Ω , and for each $k = 1, 2, \dots, r$, an open neighborhood $U_k(\xi_k)$ of ξ^k in Ξ and a continuous function $g_k : V \rightarrow U_k$ such that*
 - (a) *$U_k \cap U_{k'} = \emptyset$ if $k \neq k'$,*
 - (b) *g_k is C^1 and $g_k(e) = \xi^k$,*
 - (c) *for all $e' \in V$, $F_{e'}^{-1}(0) = \{g_k(e') : k = 1, \dots, r\}$,*
 - (d) *the economies $e' \in V$ are regular.*

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